WORKSHEET

Chapter 5: Projectile motion

Use $g \approx 9.8$ m s^{-2} and assume that air resistance is negligible.

1 A particle is projected from a horizontal surface with a speed of 80 m s^{-1} at an angle of 15° to the horizontal surface.

a Write an expression for the initial velocity of the particle.

b Calculate the speed of the particle one second after projection.

2 A ball is struck from a horizontal surface so that its speed is 40 m s^{-1} at an angle of 20° to the horizontal surface.

a Find the initial velocity of the ball.

b Write an expression for velocity of the ball as a function of time.

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3 A particle is projected into the air with an initial velocity of 10 m s^{-1} at an angle of 45°. Calculate the following:

a time of flight

b range

c greatest height reached

d position vector for the particle

e distance travelled by the particle in the first second after it was launched.

4 A ball is thrown with an initial velocity of 18 m s^{-1} at an angle of elevation of 30°. Calculate the following:

a position vector for the ball

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b greatest height the ball reaches

c time of flight

d range.

5 A particle is projected into the air with an initial velocity of 22 m s^{-1} at an angle of 60°. Calculate the following:

a time of flight

b range

c greatest height reached

d position vector for the particle

e distance travelled by the particle in the first 2 seconds after it was launched.

6 Find the greatest range possible on a horizontal plane when the speed of projection is:

a 28 m s^{-1}

b 50 m s^{-1}.

7 A particle is projected into the air from a platform 55 m above a horizontal surface with an initial velocity of 20 m s^{-1} at an angle of 40°. Calculate the following:

a time of flight

b distance between the point of projection and the point where the particle reaches the ground

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c greatest height reached

d magnitude of the velocity when the particle reaches the horizontal surface.

8 An object is thrown from the top of a cliff 30 m above the sea below. The initial velocity of the object is 32 m s^{-1} at an angle of 25°. Calculate the following:

a time of flight

b distance between the point from which the object was thrown and the point where the object reaches the ground

c greatest height reached

d magnitude of the velocity when the object reaches the ground.

9 A girl is on the top of a tower 20 m above a flat surface. She throws a stone into the air. The initial velocity of the stone is 12 m s^{-1} at an angle of 66°. Calculate the following:

a time of flight

b distance between the point where the girl throws the stone and the point where the stone reaches the ground

c greatest height reached

d magnitude of the velocity when the stone reaches the ground.

10 An object is project from a flat surface such that the position vector of the object is given by:

$\mathbf{r}(t) = 20t\mathbf{i} + (25t - 4.9t^2)\mathbf{j}$

Will the object pass over a 10 m high wall that is 94 m from the point of projection?

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Answers

1 **a** $\mathbf{v}(t) = 77.27\mathbf{i} + 20.7\mathbf{j}$

b 78 m s^{-1}

2 **a** $\mathbf{v}(0) = 37.6\mathbf{i} + 13.7\mathbf{j}$

b $\mathbf{v}(t) = 37.6\mathbf{i} + (13.7 - 9.8t)\mathbf{j}$

3 **a** 1.44 s

b 10.2 m

c 2.55 m

d $\mathbf{r}(t) = 7.1t\mathbf{i} + (7.1t - 4.9t^2)\mathbf{j}$

e 7.4 m

4 **a** $\mathbf{r}(t) = 15.6t\mathbf{i} + (9t - 4.9t^2)\mathbf{j}$

b 4.13 m

c 1.44 s

d 28.63 m

5 **a** 3.89 s

b 42.8 m

c 18.5 m

d $\mathbf{r}(t) = 11t\mathbf{i} + (19.1t - 4.9t^2)\mathbf{j}$

e 28.7 m

6 **a** 79.9 m

b 254.8 m

7 **a** 4.9 s

b 75.2 m

c 63.4 m

d 38.5 m s^{-1}

8 **a** 4.2 s

b 122.1 m

c 39.3 m

d 40.2 m s^{-1}

9 **a** 3.4 s

b 16.7 m

c 26.1 m

d 23.2 m s^{-1}

10 No, the height of the object is about 9.3 m.

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